tout vérifié

Devoir 3.2

√ = vérifié

NOTE:

$\{\{A, B, \dots, Z\} \cdot \{1, 2, \dots n\}\}$

$=$

$\{A_1, A_2, \dots, A_n, B_1, B_2, \dots B_n, Z_1, Z_2 \dots\}$

↳ Distributivité des ensembles

1.

a) $\text{card}\ 7\spadesuit$: 1 événement
$\text{card}\ U$: 52 possibilités
$P(7\spadesuit) = \boxed{\frac{1}{52}}$

b) $\text{card}\ A$: 4 événements
$\text{card}\ U$: 52 possibilités
$P(A) = \frac{4}{52} = \boxed{\frac{1}{13}}$

c) $\text{card}\ A$: $\clubsuit$ (2, 3, 4, 5, 6, 7, 8, 9, 10) (9 événements)
$\text{card}\ U$: 52 possibilités
$\boxed{P(A) = \frac{9}{52}}$

d) $\text{card}\ A$: $(\heartsuit + \spadesuit + \clubsuit + \diamondsuit) \cdot (2, 4, 6, 8, 10)$ (20 événements)
$\text{card}\ U$: 52 possibilités
$P(A) = \frac{20}{52} = \boxed{\frac{5}{13}}$

NOTE:

range (a, b)

$=$

$\{a, a+1, a+2, \dots, b-2, b-1\}$

(range(a, b) inclus a mais pas b)

2.

a) $\text{card}(U) = 3 + 2 = 5$

b) $n = \{n, n, n\}$ $P(n) = \boxed{\frac{3}{5}}$

c) $n' = R$ $P(R) = 1 - \frac{3}{5} = \boxed{\frac{2}{5}}$

d) $A = \{\}$ $P(A) = \frac{0}{6} = \boxed{0}$

3. $U = \{\text{carte} \cdot 54\}$

a) $A_J = \{J_R, J_N\}$ $P(A_J) = \frac{2}{54} = \boxed{\frac{1}{27}}$

b) $A_{JR} = \{J_N\}$ $P(A_{JR}) = \boxed{\frac{1}{54}}$

c) $A_R = \{R\heartsuit, R\diamondsuit, R\clubsuit, R\spadesuit\}$ $P(A_R) = \frac{4}{54} = \boxed{\frac{2}{27}}$

d) $A_N = \{(54/2)\ \text{cartes}\}$ (27 cartes) $P(A_N) = \frac{27}{54} = \boxed{\frac{1}{2}}$

e) $A_{<10} = \{\{A, 2, 3, 4, 5, 6, 7, 8, 9\} \cdot \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\}\}$ (36 cartes)
$P(A) = \frac{36}{54} = \boxed{\frac{2}{3}}$

Hilroy

4.

a) $U = \{Pile, Face\}$ $A = \{Pile\}$ $P(A) = \frac{card(A)}{card(U)} = \boxed{\frac{1}{2}}$

b) $U = \{1,2,3,4,5,6\}$ $A = \{3\}$ $P(A) = \boxed{\frac{1}{6}}$

c) $U = \{52 \text{ cartes}\}$ $A = \{7♠, 7♣\}$ $P(A) = \frac{2}{52} = \boxed{\frac{1}{26}}$

d) $U = \{52 \text{ cartes}\}$ $A = \{\{A,2,3,4,5,6,7,8,9,10\} \cdot \{♦, ♥, ♠, ♣\}\}$

$P(A) = \frac{40}{52} = \boxed{\frac{10}{13}}$

5. $U = \{M, A, T, Ê, M, A, T, I, Q, U, E, S\}$ $card = 13$

$A_S = \{S\}$ $card = 1$

$A_M = \{M, M\}$ $card = 2$

$A_{voy} = \{A, É, A, I, U, E\}$ $card = 6$

a) $P(A_S) = \boxed{\frac{1}{13}}$ b) $P(A_M) = \boxed{\frac{2}{13}}$ $P(A_{voy}) = \boxed{\frac{6}{13}}$

6. $A = range(221, 280)$ $U = range(1, 361)$

$card = 280 - 221 = 59$

$P(A) = \frac{card(A)}{card(U)} = \boxed{\frac{59}{360}}$

7. a) $P(7) = \frac{6}{36} = \boxed{\frac{1}{6}}$ b) $P(\neq 7) = P(\overline{7}) = 1 - \frac{1}{6} = \boxed{\frac{5}{6}}$

8. $U = \{x0, x1, x2, x3, x4, x5, x6, x7, x8, x9\}$ $A = \{xx\}$

$\boxed{P(A) = \frac{1}{10}}$